A USER MANUAL · EDITION 5.5

# RTMcompare

*Hear what Spotify hears.*
*Before Spotify hears it.*

This is a manual. Not a brochure, not a sales deck — the actual instructions for how to live inside the app.

RTMcompare is for mixing engineers chasing the next revision, mastering engineers shipping to streaming, and producers who want to know what their bounce will sound like once the platforms have had their way with it. Same tool for all three. Different starting screens depending on what you drop in.

Read it linearly or skip to the surface you're using. Chapters are short. Nobody asked for a textbook.

CHAPTER ONE

# How to install this thing.

Drag RTMcompare.app from the DMG into your Applications folder. macOS will hesitate the first time you open it — right-click → Open, click "Open" again on the Gatekeeper dialog, and from then on it's a normal app.

On first launch you meet the cover screen — a Didone wordmark, one line of instruction, one drop frame. Drag two audio files in and you're analyzing.

There's no installer wizard. No license server. No telemetry. The only network call this app makes is for opt-in DSP delivery-status fetches, and only outbound, with credentials in the macOS Keychain. Otherwise it's a closed loop on your machine.

```
Disk:    ~1.5 GB (app + bundled Python 3.11 + BS-RoFormer ckpt)
RAM:     8 GB minimum, 16 GB recommended for stem separation
Audio:   WAV, FLAC, AIFF, MP3, OGG, M4A, ADM BWF
```

# Comparing two files.

This is the surface most people opened the app to find.

Drop the reference into the left slot. Drop the file you're evaluating into the right. Pick a delivery target from the header chips — Music, Full, Bcast, Netflix, Post — that single gold pill is the lens every measurement after this gets read through. Hit Compare.

Both files are level-matched to −18 LUFS integrated before you hear a single sample. That's the whole point. Loud doesn't fake good when both files are at the same target. The version you think is better is just louder, until it isn't.

When the analysis lands, the instrument row above the tabs shows the four numbers that decide whether your file ships: LUFS-I, true peak, LRA, mono-compat. Each tab opens with a Didone verdict at the top — the headline number for that surface — then the panels you came for sit beneath.

*For: mixing engineers, mastering engineers, producers comparing their bounce to a hit.*

CHAPTER THREE

# When you only have one file.

Sometimes there's no reference. The file is the file, and you want to know whether it's ready.

Drop one audio file in. Click "Analyze Reference Only." The single-file QC pass reads the master clinically and tells you what it found:

- Click and glitch timeline, with click-to-transport jump so you can audition every defect in seconds.
- Distortion detection — clipping, ISR, harmonic — with the offending frequency band named.
- Mains hum and harmonics. 50 / 60 Hz fundamental plus the 3rd, 5th, 7th. Catches grounding issues you never thought to listen for.
- Transfer-artefact detection for analog-sourced masters: wow, flutter, DC drift, tape transport, print-through.
- Generation-loss detection. If your file came from somewhere with a lossy ancestor (a stem you got back from a producer in MP3, say), you'll see it here.
- Key, BPM, harmonic ladder. For sync pitching.
- Mono-compat as a per-band waterfall, not a scalar. You see exactly where the mono fold collapses.

*For: anyone with one file and a deadline.*

CHAPTER FOUR

# Walking an album.

Drop a folder. Get a sortable table — LUFS, true peak, LRA, ISRC, duration, sample rate, bit depth, with outliers highlighted automatically.

Click any row to drop into a single song tab. Step through the album with the arrow keys. Each song gets lazy deep analysis; results cache across rotations so you only pay the cost once.

Cohort Mode promotes any track or external file as the reference. A per-track distance heatmap across 31 bands shows which songs stray most. Sort by drift, find the outlier, fix the outlier, ship the album. That's the loop.

Save the session as a .rtmalbum.json — every analysis, every note, the A/B reference, the cohort ref, the delivery manifest state. Re-open it in a year and pick up exactly where you stopped. Useful when a label comes back asking you to remaster a 2024 EP for a vinyl press.

*For: mastering engineers shipping albums, producers auditing their catalogue.*

CHAPTER FIVE

# How Apple cancels deliveries.

Apple cancels deliveries on punctuation. "Feat." vs "feat." is enough — they auto-reject the manifest, the distributor kicks it back to you at midnight, and the release date slips.

Spotify rejects duplicate ISRCs without warning. Anyone who ships catalogue knows the feeling of a release going "live" with three of the four tracks because one ISRC collided silently with something from 2019.

The Delivery Manifest Reconciler is the surface that catches this before the email arrives. Drop the distributor's CSV or DDEX ERN 4.3 XML onto the panel inside album batch. RTMcompare three-way-diffs the audio-embedded metadata, the manifest, and the album's internal ISRC set, and surfaces every blocker:

- Title-casing drift
- ISRC collisions inside the album
- ISRC reuse from prior releases (cross-session history at ~/.rtm/isrc-history.json)
- Duration mismatches between audio and manifest
- Missing rows on either side
- P-line / C-line drift

Resolve each. Export Ship-Ready PDF and Corrected CSV. Attach both to the delivery ticket. Your distributor can re-ingest the CSV directly.

*For: label ops. Skip if you're not.*

CHAPTER SIX

# Immersive and Dolby Atmos.

Drop an ADM BWF file in. RTMcompare parses bed and objects, reads the trajectories, runs an early-warning binaural-headroom estimate (ILD downmix — no HRTF render), and QCs the stereo downmix against the immersive master.

Atmos Preflight runs the hard-checks Apple actually enforces: object count must be at most 118, LFE has to route, bed layout must be 7.1.2 or 5.1.4, sample rate 48 kHz, bit depth at least 24. Failures gate delivery. Fix before you press send.

Per-object anomaly detection flags hot, silent, static, or dark objects. Most of the time the source is a mix mistake, not artistic intent. The flag points you at the channel; you make the call about whether to fix or whether the producer really did mean to put a 6 kHz tone on object 47.

*For: immersive mix supervisors. Skippable for everyone else.*

CHAPTER SEVEN

# The two companion apps.

## *RTMprofile*

Feed it five-plus of your finished masters. RTMprofile learns the spectral and dynamic fingerprint of your work and saves it as a JSON profile. Load that profile into RTMcompare's Match tab and any new mix can be graded against your own sound, with concrete EQ moves to close the gap.

Useful if you're an engineer building a signature, or a producer who wants their EP to sound consistent.

## *RTMsend*

A VST3 / AU plugin. Sits on a bus in Wavelab, Logic, Pro Tools, Studio One, or any DAW that hosts AU/VST3.

One button sends the buffer to RTMcompare's Single, Compare-B, or Album surface. No export dialog, no render queue. ARA-aware on hosts that support it; ring-buffers the last N seconds otherwise.

Useful when you're mid-mix and need to QC something now, without breaking flow to bounce a file.

REFERENCE

# Keyboard shortcuts.

| | |
|---|---|
| Space | Play / pause the active transport |
| A · B · X | Switch A · Switch B · Flip (ABPlayer) |
| ← · → | Previous / next song (song tab) |
| 1 - 9 | Jump to tab N (Compare view) |
| M | Mono listen mode |
| S | Solo each side |
| L | Toggle loop |
| ⌘ K · / | Command palette · Song quick-switch · Search |
| ⌘ E · ⌘ ⇧ E | Export EQ (FabFilter Pro-Q text) · Apply EQ + bounce |
| ? | This shortcut legend, on screen |
| ⌘ N | New comparison (results screens) |

## Privacy.

Every analysis, every render, every metadata read happens on your machine. No audio leaves the device. The only network path the app opens is opt-in delivery-status fetching from the DSPs you've authorised — outbound, read-only, credentials in the macOS Keychain via safeStorage.